

5 Amino 1MQ Available for Research Use Only

(SubQ dosing information cannot be extrapolated because animal studies used oral dosing)

5-Amino-1MQ is likely to reduce adiposity and improve glucose regulation and other metabolic outcomes. Increased intracellular NAD⁺, adipose energy expenditure, and downstream mitochondrial effects are plausible. Long-term risks from sustained alteration of NNMT, NAD⁺ metabolism, and methyl-donor balance remain unknown. Unresolved potential risks and the absence of human safety data make this research compound unsuitable for human use without further evidence.

☠️ Possibly unsafe (mechanistic evidence)

★ ★ Probably effective (strong animal signals)

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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5-Amino-1MQ (5-amino-1-methylquinolinium) is a small-molecule NNMT inhibitor. It is likely to improve adipose metabolism, while effects on intracellular NAD⁺ biology are plausible.

The long-term safety of 5-Amino-1MQ in humans is unknown. It is possibly unsafe because mechanistic concerns from altering NAD⁺ and methylation biology outweigh limited animal tolerability signals, and human safety data are absent.

5-Amino-1MQ is likely to reduce fat mass. Increased adipose energy expenditure is a plausible contributing mechanism.-

5-Amino-1MQ may plausibly increase intracellular NAD⁺ in adipose tissue. Improved mitochondrial function is a plausible downstream effect.

Benefits

- **Likely reduction in adiposity and fat mass.** 5-Amino-1MQ is likely to reduce body weight, white-adipose mass, and adipocyte size. Increased adipose energy expenditure is a plausible contributing mechanism.
- **Plausible effects on intracellular NAD⁺ and energy metabolism.** 5-Amino-1MQ may plausibly increase intracellular NAD⁺ availability and alter downstream energy metabolism. Effects on sirtuin and mitochondrial pathways are also plausible and context-dependent.

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- **Likely improvement in glucose regulation.** 5-Amino-1MQ is likely to improve glucose tolerance and insulin sensitivity and reduce hyperinsulinemia.
- **Likely support for muscle regeneration and function.** 5-Amino-1MQ is likely to support muscle regeneration after injury and improve certain measures of muscle function in age-related muscle impairment.
- **Likely reduction in obesity-associated hepatic steatosis.** 5-Amino-1MQ is likely to reduce hepatic steatosis, liver triglyceride accumulation, and associated abnormalities in liver biomarkers.

Potential Applications

- **Obesity and reduction of adiposity**
- **Metabolic syndrome / insulin resistance**

Cautions

- **Pregnancy and breastfeeding**
- **Severe hepatic impairment**
- **Plausible interaction concerns with drugs that substantially alter methyl-donor metabolism**
- **Severe insomnia**

Reported or Possible Side Effects

- **Gastrointestinal upset (nausea, transient GI symptoms)** - anecdotally reported
- **Sleep disturbance or transient insomnia** - anecdotally reported
- **Transient restlessness, feeling hotter, or increased sweating** - anecdotally reported
- **Possible alteration of methyl-donor metabolites;** direction and clinical significance are uncertain

Drivers of Diminished Response and Mitigation Strategies

The likelihood of diminished response during sustained NNMT inhibition is unknown. Receptor desensitization and neutralizing antibodies are unlikely to be relevant because 5-Amino-1MQ is a small-molecule enzyme inhibitor rather than a receptor agonist or biologic. Downstream metabolic adaptation remains speculative.

No evidence-based strategy for preventing diminished response has been established. Limited-duration courses, cycling, or dose reduction are speculative approaches to managing a possible diminished response.

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Known Unknown Safety Risks

Human clinical data for 5-Amino-1MQ are absent. **Plausible human risks include effects from altered methylation and NAD⁺ metabolism. Uncharacterized off-target toxicity is also possible because preclinical toxicology is incomplete.**

- **Plausible alteration of liver metabolism and possible hepatotoxicity.** Because NNMT is highly expressed in the liver and participates in methyl-donor and xenobiotic metabolism, its inhibition may plausibly alter hepatic metabolic processes. Hepatotoxicity is possible.
- **Plausible alteration of one-carbon and methylation balance.** NNMT consumes SAM, so its inhibition may plausibly alter methyl-donor flux, SAM/SAH balance, and downstream DNA or histone methylation, with possible long-term effects on gene regulation.
- **Plausible tissue-dependent effects on NAD⁺ metabolism and sirtuin signaling.** By altering NAD⁺ availability, sustained NNMT inhibition may plausibly affect sirtuin activity, DNA repair, mitochondrial function, and cell survival in tissue-dependent ways. Long-term consequences remain unknown.
- **Metabolic effects after cessation are poorly characterized;** sustained benefit or rebound after discontinuation has not been established.

Biological mechanism

5Amino1MQ is a selective, membranepерmeable inhibitor of nicotinamide Nmethyltransferase (NNMT). NNMT methylates nicotinamide (NAM) using Sadenosylmethionine (SAM) to produce 1methylnicotinamide (1MNA), removing NAM from the NAD⁺ salvage pathway and consuming methyl donors. Inhibiting NNMT can increase availability of NAM and alter SAM/SAH balance, thereby altering NAD⁺ availability and cellular methyl-donor balance. Changes in NAD⁺ and methyl-donor metabolism may influence sirtuin, AMPK/PGC-1α, mitochondrial, polyamine, and epigenetic pathways. NNMT inhibition may also alter polyamine flux and epigenetic methylation states. These pathways may plausibly contribute to reduced adipocyte hypertrophy and improved insulin sensitivity. Increased energy expenditure is also a plausible contributing mechanism.

Evidence Summary

5-Amino-1MQ is likely to reduce adiposity, improve glucose regulation, reduce obesity-associated hepatic steatosis, and support certain muscle-regenerative and functional outcomes. Increased intracellular NAD⁺, adipose energy expenditure, and downstream mitochondrial effects are plausible. Long-term safety remains unknown; pathway-related risks are plausible, and uncharacterized off-target risks are possible.

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Criterion	Mechanistic / preclinical	Human data	Implication
Target / mechanism	NNMT inhibition → preserves nicotinamide for NAD ⁺ salvage; alters SAM/methylation.	Minimal peer-reviewed human data.	Strong target engagement in preclinical models; systemic consequences remain incompletely characterized.
Adipose/weight effects	Robust mouse VAT loss, increased energy expenditure.	Small case reports / clinic summaries only.	Plausible translation; human proof lacking.
NAD ⁺ /mitochondria	Cellular and animal NAD ⁺ increases, improved mitochondrial markers.	No robust human RCTs.	Mechanistically strong; human confirmation needed.
Safety	Animal tolerability reported; mechanism alters methylation (plausible risk).	Human long-term safety data absent.	Potential systemic/epigenetic risks; monitor closely.

Conclusions

5-Amino-1MQ is likely to improve adiposity and several metabolic outcomes. Increased intracellular NAD⁺, adipose energy expenditure, and downstream mitochondrial effects are plausible. Effective human dosing and long-term safety remain unknown.

References

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